

MRA Assumptions Exploration

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Model building

Linear regression relies on several assumptions. In this project I explore what happens when those assumptions are violated, and more specifically **which estimated quantities change the most** when each violation is introduced in isolation. I focus on:

1. **Linearity**: the conditional mean of the outcome is a linear function of the predictors.
2. **Normality of errors**: the residuals are normally distributed around zero.
3. **Homoscedasticity**: the error variance is constant across fitted values.
4. **Independence of errors**: residuals are uncorrelated across observations.

To study this, I simulate a “perfect” model first and then alter one assumption at a time. For each scenario, I compare the resulting:

- intercept
- regression coefficients for working memory and IQ
- standard errors
- residual standard error
- R^2

I also keep one representative dataset from each scenario to visualize how the assumption is violated.

The data generating model is:

$$\text{math_test}_i = 22 + 0.2 \text{IQ}_i + 1.1 \text{WM}_i + \varepsilon_i, \quad i = 1, \dots, n.$$

with predictors generated as

$$\text{IQ}_i \stackrel{iid}{\sim} \mathcal{N}(100, 15^2), \quad \text{WM}_i \stackrel{iid}{\sim} \mathcal{N}(26, 8^2),$$

and under the benchmark model,

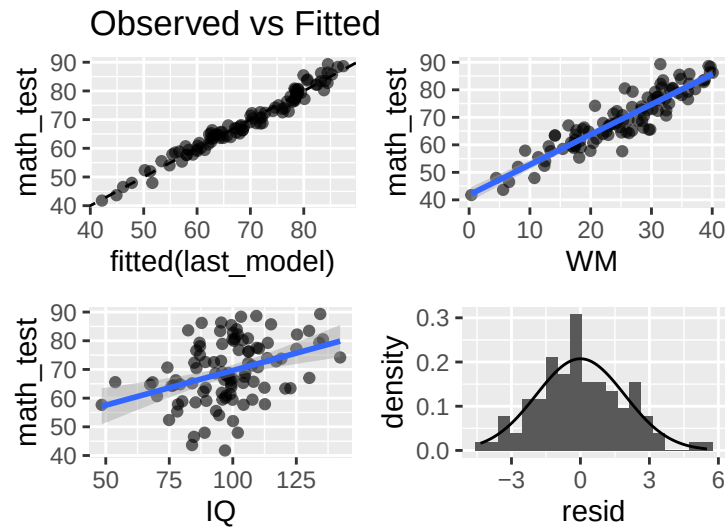
$$\varepsilon_i \stackrel{iid}{\sim} \mathcal{N}(0, 2^2).$$

So under the ideal case,

$$\text{math_test}_i \mid (\text{IQ}_i, \text{WM}_i) \sim \mathcal{N}(22 + 0.2 \text{IQ}_i + 1.1 \text{WM}_i, 2^2).$$

Perfect model

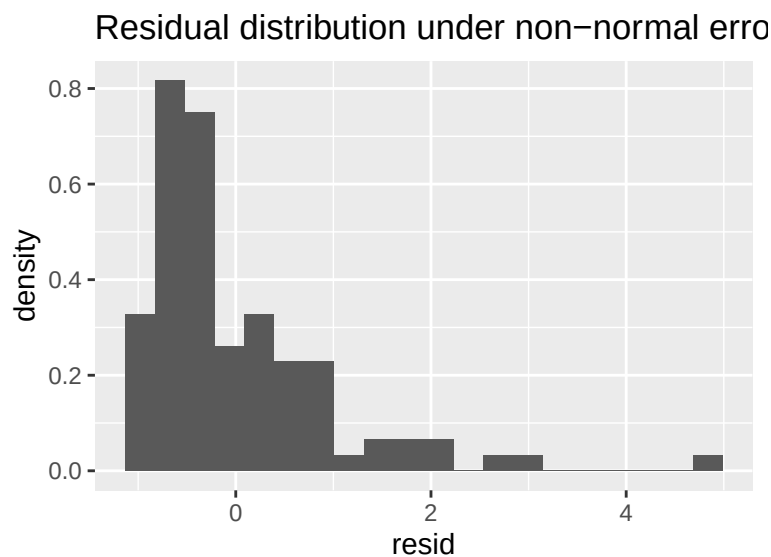
Visualizations for the perfect model



The plots do not show violations of the key assumptions of multiple regression. The dependent variable and the independent variables have a linear relationship, the errors are approximately normal, and have constant variance across the fitted values.

Violation of Normality

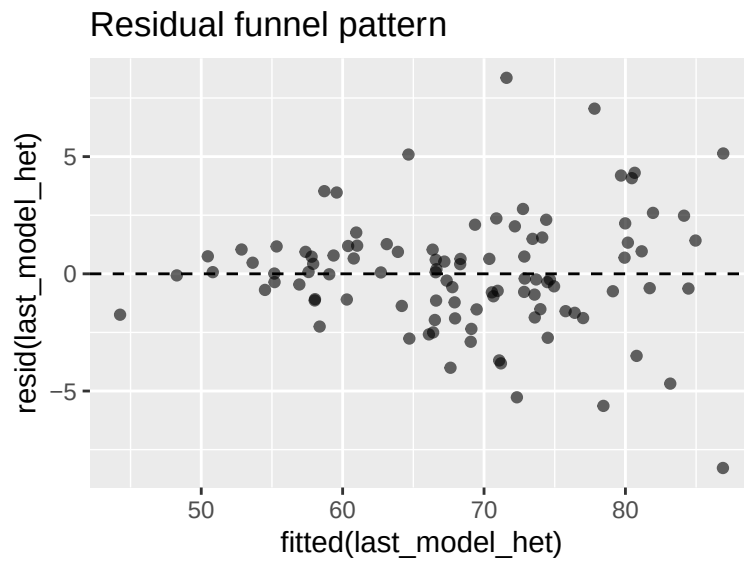
To violate the normality assumption, I generate skewed errors using an exponential distribution.



Even though the residual distribution is clearly skewed, the regression coefficients remain close to the true values. This illustrates that **normality mainly affects inference rather than the coefficient estimates themselves**.

Violation of Homoscedasticity

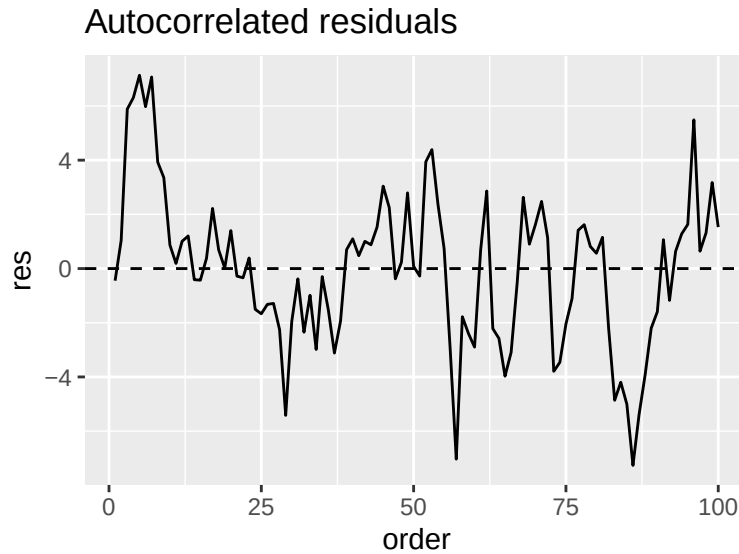
Now the variance of the errors increases with working memory.



The funnel shape indicates increasing variance. Again, the coefficients remain largely unbiased, but **standard errors are affected because the model assumes constant variance**.

Violation of Independence

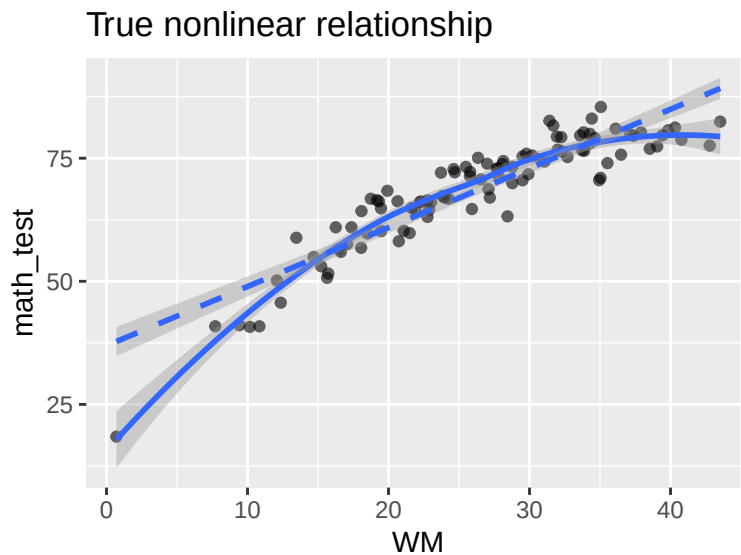
Autocorrelation is introduced by making each residual depend on the previous residual.



Residuals now show a clear pattern over observations, violating independence.

Violation of Linearity

Finally, I introduce a quadratic relationship with working memory.



Here the linear regression tries to approximate a curved relationship with a straight line. This leads to **biased slope estimates**, making this assumption violation particularly problematic.

Overall Conclusion

Table 1: Average estimated parameters across 3,000 simulations per scenario (true values: intercept = 22, b_WM = 1.1, b_IQ = 0.2)

Scenario	Intercept	Coef (WM)	Coef (IQ)	SE (WM)	SE (IQ)	p (WM)	p (IQ)	Residual SE	R-sq
Perfect Model	21.986	1.100	0.2	0.025	0.014	0	0	1.998	0.956
Non-normal Errors	21.993	1.100	0.2	0.013	0.007	0	0	0.990	0.989
Heteroscedastic Errors	21.996	1.100	0.2	0.032	0.017	0	0	2.548	0.930
Autocorrelated Errors	21.993	1.100	0.2	0.033	0.017	0	0	2.560	0.929
Nonlinear Relationship	20.173	1.099	0.2	0.042	0.022	0	0	3.308	0.888

The simulation illustrates that not all regression assumptions affect the model in the same way.

Violations of **normality**, **homoscedasticity**, and **independence** mainly affect the **uncertainty estimates** of the model. The regression coefficients themselves remain close to their true values across all three scenarios, indicating that OLS estimation is relatively robust to these violations in terms of bias. However, the standard errors become unreliable: heteroscedasticity inflates them for the affected predictor (WM), while autocorrelation leads to underestimated standard errors, making the model appear more precise than it actually is. Non-normal errors have the least visible impact on the standard errors, consistent with the central limit theorem providing some protection at $n = 100$.

In contrast, violations of **linearity** directly distort the estimated regression coefficients. Because the model forces a straight line through a curved relationship, the WM slope is pulled away from the true value — this is not a matter of noisy estimation but systematic bias. No amount of additional data would fix this; the model is simply misspecified.

From an applied perspective, this means the consequences of each violation differ in kind, not just degree. Normality, homoscedasticity, and independence violations call for corrections to inference (e.g. robust standard errors, GLS), while a linearity violation requires reconsidering the model structure itself. **Among all the assumptions explored here, misspecification of the mean function is the most damaging because it compromises the coefficients that are typically the primary target of interpretation.**